# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 6 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 2

**Full Syllabus (Chapters 1–10)** | RBSE Pattern | Session 2026–27

**Time: 3 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 80**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 20 × 1 = 20

*Choose the correct option and write it in your answer book.*

**1.** The sum of the triangular numbers 10 and 15 is: 1

**(a)** 25 **(b)** 24 **(c)** 29 **(d)** 21

**2.** Angles $\angle AOB = 35^\circ$ and $\angle BOC = 40^\circ$ are adjacent (they share the arm OB, and lie on opposite sides of it). Then $\angle AOC$ is: 1

**(a)** 75 **(b)** 105 **(c)** 5 **(d)** 85

**3.** In the hailstone (Collatz) sequence, the number after 18 is: 1

**(a)** 9 **(b)** 10 **(c)** 36 **(d)** 55

**4.** Using the table, how many more are there of Banana than Guava? 1

| Item | Number |
|---|---|
| Guava | 20 |
| Banana | 40 |
| Grapes | 35 |

**(a)** 20 **(b)** 25 **(c)** 15 **(d)** 60

**5.** Which of these is a pair of twin primes? 1

**(a)** 13 and 17 **(b)** 11 and 13 **(c)** 19 and 23 **(d)** 7 and 11

**6.** A square has area 64 $m^2$. Its perimeter (in m) is: 1

**(a)** 48 **(b)** 40 **(c)** 24 **(d)** 32

**7.** Which is the greatest of $\frac{3}{5}$, $\frac{5}{8}$, $\frac{7}{12}$, $\frac{2}{3}$? 1

**(a)** $\frac{3}{5}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{7}{12}$ **(d)** $\frac{5}{8}$

**8.** The diagonals of a rectangle are each 10 cm long and meet at O. The length OA (cm) is: 1

**(a)** 4 **(b)** 6 **(c)** 10 **(d)** 5

**9.** The mirror image of the letter A in a vertical mirror to its right looks like: 1

**(a)** a different letter **(b)** the same letter
**(c)** an upside-down letter **(d)** a reversed letter

**10.** On the number line, the integer -6 lies 6 units to the left of 0. The integer is: 1

**(a)** -6 **(b)** -5 **(c)** 6 **(d)** -7

**11.** What is the sum of the first 15 natural numbers $1 + 2 + 3 + \cdots + 15$? 1

**(a)** 225 **(b)** 135 **(c)** 240 **(d)** 120

**12.** You face North and turn clockwise through 90 degrees. What further clockwise turn (in degrees) brings you back to facing North? 1

**(a)** 90 **(b)** 270 **(c)** 360 **(d)** 180

**13.** The number of steps to reach a palindrome by repeatedly reversing and adding, starting from 57: 1

**(a)** 4 **(b)** 3 **(c)** 2 **(d)** 1

**14.** The total number of students in the table is: 1

| Sport | Students |
|---|---|
| Hockey | 30 |
| Tennis | 20 |
| Cricket | 25 |

**(a)** 55 **(b)** 85 **(c)** 70 **(d)** 75

**15.** The number of prime numbers between 10 and 30 is: 1

**(a)** 8 **(b)** 7 **(c)** 6 **(d)** 5

**16.** The area of a rectangle is 144 $cm^2$ and its length is 16 cm. Its breadth (in cm) is: 1

**(a)** 7 **(b)** 9 **(c)** 18 **(d)** 25

**17.** $1\frac{1}{2} + 2\frac{1}{3} =$ 1

**(a)** $3\frac{1}{2}$ **(b)** $3\frac{1}{6}$ **(c)** $3\frac{5}{6}$ **(d)** $3\frac{2}{5}$

**18.** In rectangle ABCD, diagonals meet at O and $\angle AOB = 60°$. Then $\angle OAB$ is (degrees): 1

**(a)** 60 **(b)** 30 **(c)** 120 **(d)** 70

**19.** A figure coincides with itself after turns of 90, 180, 270, ... degrees, up to a full turn of 360°. Its order of rotational symmetry is: 1

**(a)** 4 **(b)** 6 **(c)** 5 **(d)** 3

**20.** The number of integers between -5 and 4 is: 1

**(a)** 7 **(b)** 10 **(c)** 8 **(d)** 9

## SECTION — B (Very Short Answer Type) 6 × 2 = 12

*Answer in about 20–30 words each.*

**21.** Add $\frac{1}{4}$ and $\frac{1}{6}$. 2

**22.** A circle has radius 6 cm. A point P is 4 cm from its centre. Is P inside, on or outside the circle? Give the reason. 2

**23.** Draw (describe) all lines of symmetry of a rectangle (not square). 2

**24.** Show $5 + (-3)$ using tokens (positive tokens and negative tokens). Find the value. 2

**25.** How many lines are needed to join all pairs among 8 points (a complete graph)? 2

**26.** What is the smaller angle between the hands of a clock at 3 o'clock? 2

## SECTION — C (Short Answer Type) 6 × 3 = 18

*Answer in about 40–60 words each.*

**27.** A regular polygon has rotational symmetry of angle 40°. How many sides does it have, and how many lines of symmetry? 3

**28.** Evaluate $8 + (-12) + 5$. 3

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**29.** Write the sum $1^3 + 2^3 + \cdots + 3^3$ and show it equals the square of $1 + 2 + \cdots + 3$. 3

**30.** Using a protractor, draw an angle of $40^\circ$ and name its type. 3

**31.** Find the number of steps taken by the Collatz sequence starting at 13 to reach 1 and the largest number in it. 3

OR

A scale 1 cm = 10 units is used. Bars are of height 4 cm, 7 cm and 5 cm. Find the values, and the difference between the largest and the smallest. 3

**32.** Draw a factor tree (describe) for 96 and write its prime factorisation. 3

OR

Find the area of a right-angled triangle with sides 6 cm, 8 cm and hypotenuse 10 cm, and its perimeter. 3

## SECTION — D (Long Answer Type) — 6 × 5 = 30

*Answer in about 100–150 words each.*

**33.** A submarine is at -80 m. It rises 30 m, then dives 25 m. Find its final depth and the total vertical distance travelled. Is it below sea level? 5

**34.** In the Virahanka sequence $1, 2, 3, 5, 8, \ldots$ the 5th and 6th terms are 8 and 13. Find the next three terms and add the first 5 terms. 5

**35.** The hour hand of a clock moves clockwise from 1 o'clock to 4 o'clock. Find the angle it turns through. In the same time, how many minutes pass and what angle does the minute hand turn through? Name the type of the hour hand's turn. 5

**36.** Show with working that adding 121 and 232 (both palindromes) gives 353. Is the sum a palindrome? Make a palindrome using the reverse-and-add process starting from 29. 5

OR

Data of 14 students' number of siblings: 5, 3, 5, 5, 5, 1, 2, 2, 2, 4, 0, 3, 3, 4. Prepare a frequency table, and find the most common number of siblings and how many students have more than 2 siblings. 5

**37.** A number is divisible by both 3 and 4. Show that it is divisible by 12 when 3 and 4 are co-prime. Give an example with 60. 5

OR

A rectangular garden is 20 m by 15 m. A path 2 m wide runs around it on the outside. Find the area of the path. 5

**38.** A tank holds $2\frac{1}{2}$ litres of water. $1\frac{1}{3}$ litres more are poured in and $\frac{1}{4}$ litre is used. Find the water left. 5

OR

A circle has centre O and radius 6 cm. Points P, Q, R are at distances 4 cm, 6 cm and 9 cm from O. State whether each lies inside, on or outside the circle, and how many lie inside or on the circle. 5

## END OF QUESTION PAPER — Total: 80 Marks